Question ID: 9fc31513

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in one variable	Easy

Question

The perimeter of an isosceles triangle is 36 feet. Each of the two congruent sides of the triangle has a length of 10 feet. What is the length, in feet, of the third side?

Answer

A. 10

B. 12

C. 16

D. 18

Correct Answer: C

Rationale

Choice C is correct. It's given that the perimeter of an isosceles triangle is 36 feet and that each of the two congruent sides has a length of 10 feet. The perimeter of a triangle is the sum of the lengths of its three sides. The equation $10 + 10 + x = 36$ can be used to represent this situation, where x is the length, in feet, of the third side. Combining like terms on the left-hand side of this equation yields $20 + x = 36$. Subtracting 20 from each side of this equation yields $x = 16$. Therefore, the length, in feet, of the third side is 16.

Choice A is incorrect. This would be the length, in feet, of the third side if the perimeter was 30 feet, not 36 feet.

Choice B is incorrect. This would be the length, in feet, of the third side if the perimeter was 32 feet, not 36 feet.

Choice D is incorrect. This would be the length, in feet, of the third side if the perimeter was 38 feet, not 36 feet.